

YUANZAN

qPCR Series
VQ-200



I Dedicated to premium quality I

Shanghai Yuanzan Life Science Co., Ltd
www.yuanzan.com



10⁺
Years
Industry Exp

300⁺
Product Series

28,000⁺
m²
Manufacturing Base

200⁺
Professionals

800⁺
Clients

30⁺
Patents Worldwide

COMPANY OVERVIEW

YUANZAN Life Science Co., Ltd. is a precision medicine enterprise led by Academician Ding Jian.

Our R&D and manufacturing subsidiary, Beijing Zhiyu Biotechnology Co., Ltd. was founded in 2017.

VQ-200 Series

A Truly Open, High-Precision
qPCR System

- ▶ Not locked to proprietary consumables.
 - ▶ Not tied to an expensive external computer.
-
- ▶ Not trapped in a closed analysis system.



**Just accurate, open,
and affordable qPCR**

Why 'Open' Matters in qPCR

The VQ-200 is different

Most qPCR systems on the market today lock you in — proprietary consumables, bundled computers, closed analysis algorithms.



Consumable Freedom

Compatible with any brand of 0.1/0.2 mL tubes, 8-strip tubes, and 96-well plates.

No proprietary plastics.

No premium consumables.



No Bundled Computer

Built-in 9" touch screen and embedded computer.

No separate PC needed.

VQ-200 offers USB/RS485

for custom integration.



Open Analysis Algorithm

Not a black box. Acquisition parameters and algorithms can be customized for specific reagent kits. Custom reports. LIMS integration.



Automation Ready

Motorized lid, auto-loading chamber, USB/RS485.

Ready for robotic integration and dark-lab environments.



► VQ-200 Series

◇ 96-well, 4-channel real-time PCR system. Performance that competes with the best — at a price that makes sense.

► Models

- ◇ VQ-200A: Standalone with embedded computer, 9" touch screen, backup battery.
- ◇ VQ-200B: External computer control. For OEM/integration projects.

► Key Specifications

Parameter	VQ-200	VQ-100
Dimensions	Semiconductor Peltier Module	Semiconductor Peltier Module
Weight	Module Temperature Control	Module Temperature Control
Display	4.0~99.0°C	4.0~99.0°C
Wells	≥ 7°C (5°C) / S	≥ 7°C (5°C) / S
Consumables	≤ ± 0.1°C	≤ ± 0.15°C
Backup Battery	≤ ± 0.1°C	≤ ± 0.1°C
Connectivity	50.0~110.0°C	50.0~110.0°C

► Thermal Performance

Parameter	VQ-200	Industry Std
Accuracy	±0.1°C	≤ 0.2°C
Uniformity	±0.15°C	± 1°C
Max Heating	≥ 7°C/s	≥ 4°C/s
Max Cooling	≥ 5°C/s	≥ 2.0°C/s
Range	4.0 – 99.0°C	—

► Optical Detection

Channel	Dyes	Ex/Em (nm)
1	FAM, SYBR Green	470 / 525
2	HEX, JOE, VIC, TET	525 / 565
3	CY5, Texas Red	624 / 675
4	ROX	571 / 615
Source	High-power white LED (450~750 nm)	
Detector	CMOS camera (500~800 nm)	

► Performance

Parameter	Value
Dynamic Range	1 – 10 ¹⁰ copies
Sensitivity	Single copy
Linearity (r)	≥ 0.99
Reproducibility (Ct CV)	≤ 1%
Volume	5 – 100 µL
Methods	Qualitative, Absolute/RQ, Melt Curve

Product Features

- ◇ The high-brightness white LED provides an efficient and stable light source.
- ◇ A high-precision multi-channel temperature calibration system ensures temperature control accuracy and uniformity across all modules.
- ◇ A dual-channel PID algorithm guarantees the precision and accuracy of temperature control.
- ◇ The optical filter band selection algorithm minimizes fluorescence crosstalk between channels.
- ◇ Compact and lightweight benchtop device, with small size (24cm*38cm*38cm) and light weight<18kg)
- ◇ Supports various analysis methods, including qualitative analysis, quantitative analysis and melt curve analysis

Product Specifications

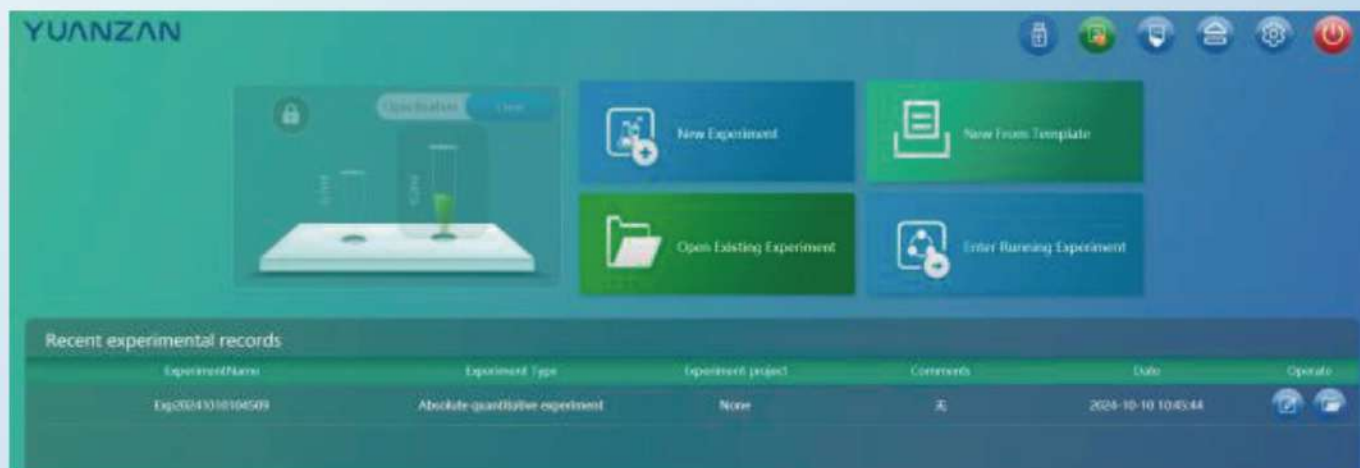
Performance Parameters	
Module Wells	96 wells
Reaction Volume	10-100µL
Consumables	0.1/0.2 mL PCR tubes, 8-strip tubes, 96-well plates
Dynamic Range	1-10 ¹⁰ copies
Detection Sensitivity	1 copy
Melting Curve Resolution	0.1°C
Multiplex Fluorescence Detection Channels & Reporter Dyes	1: FAM, SYBR-Green
	2: HEX, JOE, VIC, TET
	3: CY5, Texas-Red
	4: ROX
	5: Custom
	6: Custom

Thermal Cycling System	
Temperature Control Method	Semiconductor thermoelectric modules
Temperature Control Mode	Block control
Temperature Range	4.0 - 99.0°C
Max Heating/Cooling Rate	≥ 7°C (5°C) / S
Temperature Uniformity	≤ ± 0.15°C
Temperature Accuracy	≤ ± 0.1°C
Lid Temperature Range	50.0 ~ 110.0°C

Optical Detection System	
Light Source	LED
Detector	CMOS
Excitation Wavelength	450-750 nm
Emission Wavelength	500-800 nm
Fluorescence Reproducibility	CV ≤ 0.75%
Fluorescence Linearity	r ≥ 0.99

Analysis Software	
Software Applications	Qualitative analysis, melting curve, etc.
Scan Mode	Full-plate scanning
Data Export Formats	excel, pdf
Custom Report	Supported

Six Reasons to Choose VQ-200



Independent performance validation

1 Performance That Competes with the Best

Independently tested:

- SYBR Green reproducibility CV = 0.17% — beats Brand A (0.41%) and Brand B (1.87%)
- Clinical HBV correlation: $R^2 = 0.99$, systematic error 1.7%
- Temp accuracy $\pm 0.1^\circ\text{C}$ — 2× better than industry standard

2 Use Any Consumables, Any Brand

Compatible with all standard 0.1/0.2mL tubes, strips, and plates from any manufacturer. No proprietary consumables. Your existing lab inventory works.

3 No Computer Required

VQ-100A: built-in computer and 9" touch screen. Turn it on and run. No external PC to purchase or maintain. Software pre-installed.

4 Automation-Ready

Motorized lid, auto chamber, RS485/USB. Ready for robotic integration and dark-lab workstations. No expensive add-ons.

5 Total Cost of Ownership You Control

No mandatory service contracts. No consumables lock-in. Domestically sourced components keep both purchase price and operating costs predictable.

6 Your Algorithms, Your Reports

Customizable acquisition parameters and analysis algorithms for specific reagent kits. Custom report formats. LIMS-compatible data export (Excel, PDF, CSV).

► Performance Validation

◇ Independent testing vs. leading international platforms.

Assay	VQ-200	Brand A	Brand B
SYBR (DNA) CV	0.17%	0.41%	1.87%
Cy5 (RNA) CV	0.32%	0.62%	1.07%
Amplification Eff	100%	99.98%	99.99%

SYBR Green CV 0.17% — significantly outperforms both competitors.

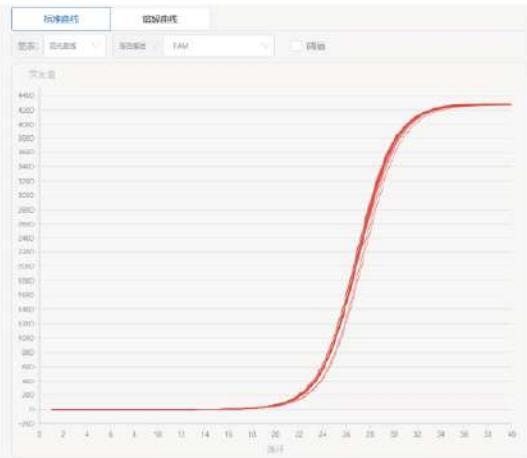
► Clinical Correlation

- ◇ HBV (20+/10- samples):
- Error vs. reference: 1.7% (acceptance: 5%)
 - $R^2 = 0.99$
- Influenza A (20+/10- samples):
- Positive agreement: 100%

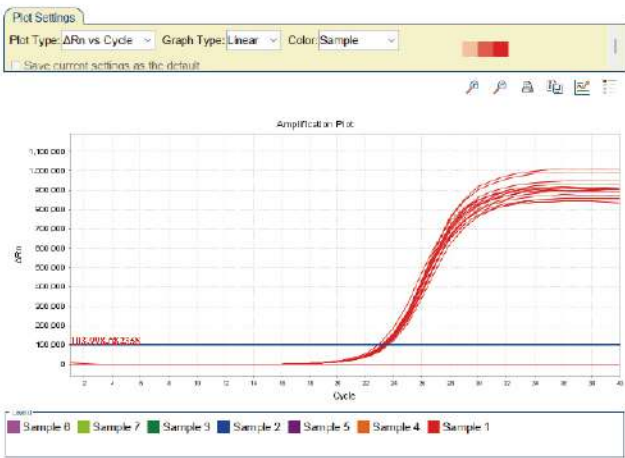
► Performance Demonstration

◇ High Repeatability

Model	VQ-200				A certain imported brand			
Repeata	A	B	C	D	A	B	C	D
1	22.84	23.01	23.45	22.99	23	23.35	-	23.30
2	23.08	22.87	23.05	22.99	23	23.34	22.97	23.40
3	23.11	23.06	23.12	23.53	23.33	23.55	23.28	23.71
4	23.05	22.97	23.02	22.80	23.23	23.45	23.61	23.55
	AVE	ST	CV		AVE	ST	CV	
	23.37	0.18	0.01		23.06	0.19	0.01	



VQ-200

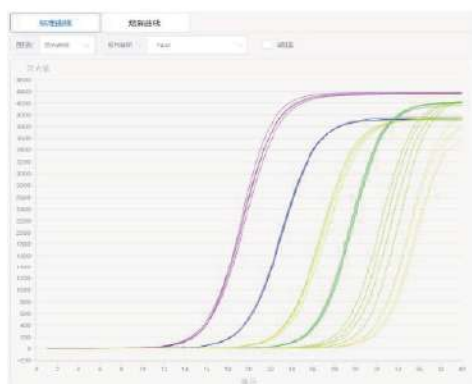


A leading imported brand

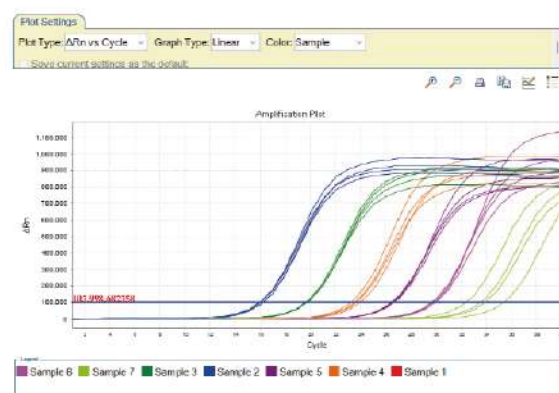
► Performance Demonstration

◇ High Resolution

Model		VQ-200							A leading imported brand						
Sample	Concentration	Rep1	Rep2	Rep3	Rep4	Rep5	AVE	Ct value difference 1 ± 0.2	Rep1	Rep2	Rep3	Rep4	Rep5	AVE	Ct value difference 1 ± 0.2
1	1ng/ μ L	20.41	20.31	20.37	20.37	20.33	20.358	/	21.9	21.82	21.87	21.87	21.87	21.866	/
2	0.5ng/ μ L	21.52	21.43	21.47	21.42	21.478	21.478	1.12	22.98	22.96	22.83	22.93	23.02	22.944	1.078
3	0.25ng/ μ L	22.57	22.48	22.38	22.35	22.464	22.464	0.986	24.16	24.06	24.02	23.99	24.06	24.058	1.114
4	0.125ng/ μ L	23.63	23.69	23.87	23.76	23.736	23.736	1.272	25.56	25.2	25.15	25.18	25.6	25.338	1.28
5	0.0625ng/ μ L	24.63	24.97	24.91	24.72	24.83	24.83	1.094	26.51	26.24	26.35	26.28	26.28	26.332	0.994



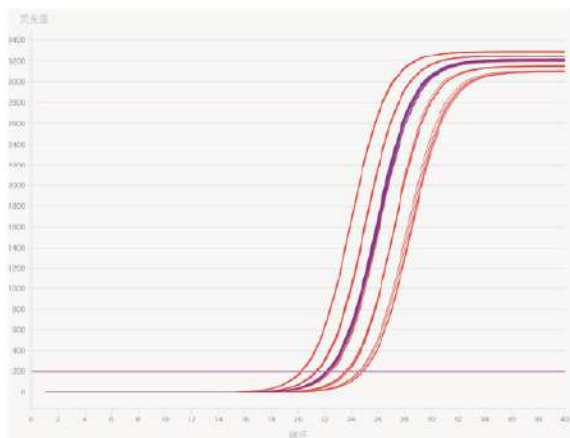
VQ-200



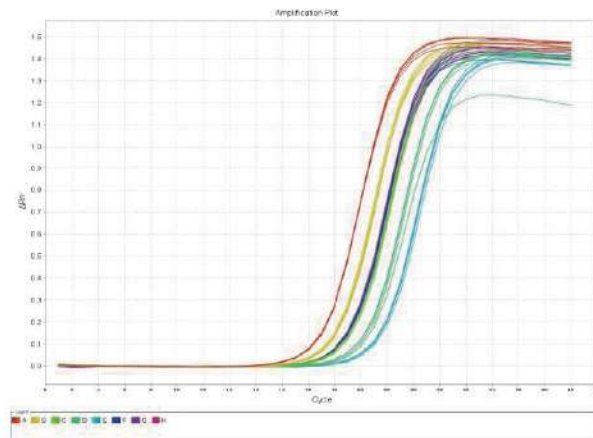
A leading imported brand

◇ High Accuracy

Model		VQ-200							A leading imported brand						
Gradient		Rep1	Rep2	Rep3	Rep4	AVE	ST	CV	Rep5	Rep6	Rep7	Rep8	AVE	ST	CV
1 E6		15.67	15.63	15.43	15.66	15.60	0.112952792	0.72%	16.01	16.27	16.26	16.09	16.15671396	0.128683576	0.80%
1 E5		19.29	19.29	19.16	19.3	19.26	0.066833126	0.35%	19.77	19.81	19.76	19.68	19.75334787	0.056533259	0.29%
1 E4		23.4	23.09	23.12	23.04	23.16	0.161735381	0.70%	23.57	23.68	24.01	23.26	23.63282347	0.307387399	1.30%
1 E3		25.92	25.85	26.03	25.83	25.91	0.090323493	0.35%	26.72	26.58	26.67	26.51	26.61919451	0.093538696	0.35%
1 E2		29.7	28.67	29.22	30.57	29.54	0.805357064	2.73%	30.01	29.80	29.98	29.74	29.88468313	0.132597692	0.44%
1 E1		32.4	31.56	32.19	31.68	31.96	0.402026119	1.26%	32.43	33.61	35.46	33.98	33.87345791	1.24882062	0.37%



VQ-200



A leading imported brand

Simple 3-Step Workflow

Step 1: Prepare



Prepare PCR mix, load into plate or tubes.

Step 2: Load



Open chamber, place consumables, close. Auto lid.

Step 3: Run



Select template, tap "Run." System handles the rest.



Software

- Template-based one-click setup
- Real-time monitoring
- Standard/melt curve, Ct analysis
- Custom reports, LIMS export

Service & Support



Customer Support

- Regular visit program
- 24-hour technical response
- Maintenance & repair

Customization

- Reagent-specific optimization
- Custom reports
- Software customization
- On-site training



Certifications

- Medical Device Registration
- 30+ patents
- GB4793.1, GB/T18268.1/26

► Application area



Medical diagnosis and screening



Gene expression research



Food and environmental monitoring



Plant and microbiological research



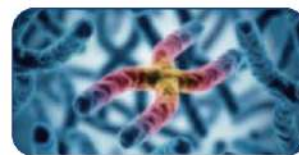
Pathogen detection



Epidemiological research applications



Drug development and evaluation



Genetic disease analysis

► Published Research Collection

- ◇ Our qPCR instruments have been used in publications in journals such as PNAS and Cell Reports by universities

Phosphorylation of LZTS2 by PLK1 activates the Wnt pathway

Xiao Liu¹, Zhi Chen², Benjing Yu¹, Zhiyong Zhou³

ARTICLE IN PRESS

PMID: 35485107 DOI: 10.1093/nci/nkz117

Cell article

Abstract

Lung adenocarcinoma (LUAD) is responsible for nearly half of lung cancer cases. In patients with LUAD, the expression of LZTS2 is downregulated. However, the precise mechanism by which LZTS2 promotes tumorigenesis in LUAD is not clear. In this study, we demonstrate the phosphorylation of LZTS2 by PLK1, which promotes the interaction between LZTS2 and β-catenin, resulting in an increase of β-catenin and the activation of the Wnt pathway. Additionally, we observe that the phosphorylation of LZTS2 by PLK1 is required for the inhibition of β-catenin-dependent transcription of the Wnt target genes. Our results uncover the phosphorylation of LZTS2 by PLK1 as a potential therapeutic target for LUAD.

SPOP point mutations regulate substrate preference and affect its function

Nanxin Deng¹, Wenxiao Ding², Kailin Ma³, Meizhao Zhen⁴, Li Sun⁵, Zhiyong Zhou⁶

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Cell article

Abstract

The adaptor SPOP recruits substrates to CUL3 E3 ligase for ubiquitination and degradation. Structurally, SPOP harbors a MATH domain for substrate recognition and a BTB domain for recruiting CUL3. Reported point mutations always occur in SPOPs MATH domain and affect its substrate preference. In this study, we demonstrate that SPOP point mutations affect its substrate preference and affect its function. We observe that the phosphorylation of SPOP by PLK1 is required for the inhibition of β-catenin-dependent transcription of the Wnt target genes. Our results uncover the phosphorylation of SPOP by PLK1 as a potential therapeutic target for LUAD.

The AAA-ATPase Ter94 regulates wing size in Drosophila by suppressing the Hippo pathway

Mingming Li¹, Wenxiao Ding², Nanxin Deng³, Wenxiao Ding⁴, Qiqin Lu⁵, Zhiyong Zhou⁶

ARTICLE IN PRESS

PMID: 35485107 DOI: 10.1093/nci/nkz117

Cell article

Abstract

Insect wing development is a fascinating and intricate process that involves the regulation of cell proliferation and apoptosis. In this study, we find that Ter94, an AAA-ATPase, is essential for proper wing size development in Drosophila. Loss of Ter94 results in increased apoptosis, which can be rescued by inhibiting the Hippo pathway. Biochemical assays reveal that Ter94 specifically binds to MstA, a critical upstream component of the Hippo pathway. This interaction prevents the formation of the TSC1 complex, ultimately leading to the inactivation of the Hippo pathway and promoting wing development. Finally, we show that MCTF, the human homolog of Ter94, is able to substitute for Ter94 in regulating wing size, underscoring their functional conservation. In conclusion, Ter94 plays a positive role in regulating wing size by interacting with the TSC1 complex, in the suppression of the Hippo pathway.

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Oxidative stress reprograms the transcriptional coactivator Yki to suppress cell proliferation

Qiqin Lu¹, Wenxiao Ding², Nanxin Deng³, Wenxiao Ding⁴, Qiqin Lu⁵, Zhiyong Zhou⁶

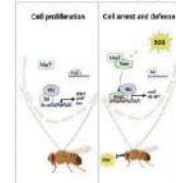
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Cell article

Abstract

The transcriptional coactivator Yki (Yki) regulates organ size by promoting cell proliferation. It is a downstream effector of the Hippo pathway, which is a conserved signaling pathway. In this study, we show that oxidative stress reprograms the transcriptional coactivator Yki to suppress cell proliferation. We observe that the phosphorylation of Yki by the Hippo pathway is required for its activation. Under oxidative stress, Yki is phosphorylated by the Hippo pathway, leading to its inactivation and suppression of cell proliferation. Our results uncover the phosphorylation of Yki by the Hippo pathway as a potential therapeutic target for LUAD.



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► Academic & Corporate Partners

- ◇ We have collaborated with nearly a hundred universities and research institutions both domestic and internationally, listed in no particular order



※ YUANZAN provides comprehensive life science solutions.

Welcome to visit our official website for more product information.

YUANZAN



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